

ASSEMBLY INSTRUCTIONS FOR SCH1000 SERIES

Technical Note

CONFIDENTIAL

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1 Objective

This document provides required and recommended items and guidelines for Printed Circuit Board (PCB) design and assembly of Murata's SCH1000 family components. It should be emphasized that this document serves only as a design guideline to help to develop the optimal assembly conditions. It is essential that users also use their own manufacturing practices and experience to be able to fulfill the needs of varying end-use applications.

Required items must be fulfilled by the user to ensure component operation in application. Regarding recommended items, not following the recommendations might result in deviations from the specified performance. Required items are marked with label (*Requirement*) and it refers to the previous sentence when it is followed by a period. When the label (*Requirement*) is after a period mark, it refers to the whole section. Same notation applies to Recommended items with the label (*Recommendation*). User is responsible to evaluate and qualify implemented application design options in the final product.

This document can be continuously updated in future. Please refer to the latest document.

2 Small Outline Integrated Circuit (SOIC) package

Figure 1 shows SCH1000 family SOIC-24 package. The package is lead-free, pick-and-place mountable, and meets industry standard lead-free soldering processes.



Figure 1 SOIC-24 package

The outline and dimensions for the SOIC package are shown in **Figure 2**.

Figure 2 Outline of SOIC package. All dimensions are in millimeters. All angles are in degrees. Tolerances unless otherwise specified according to ISO2768-f. A sample part number for reference only. A 3D model of the component is available upon request.

4 Tape and reel specifications

Packing tape dimensions are presented in **Figure 3**. The reel dimensions, unreeling direction and component polarity on tape are presented in **Figure 5** and **Table 1** below.

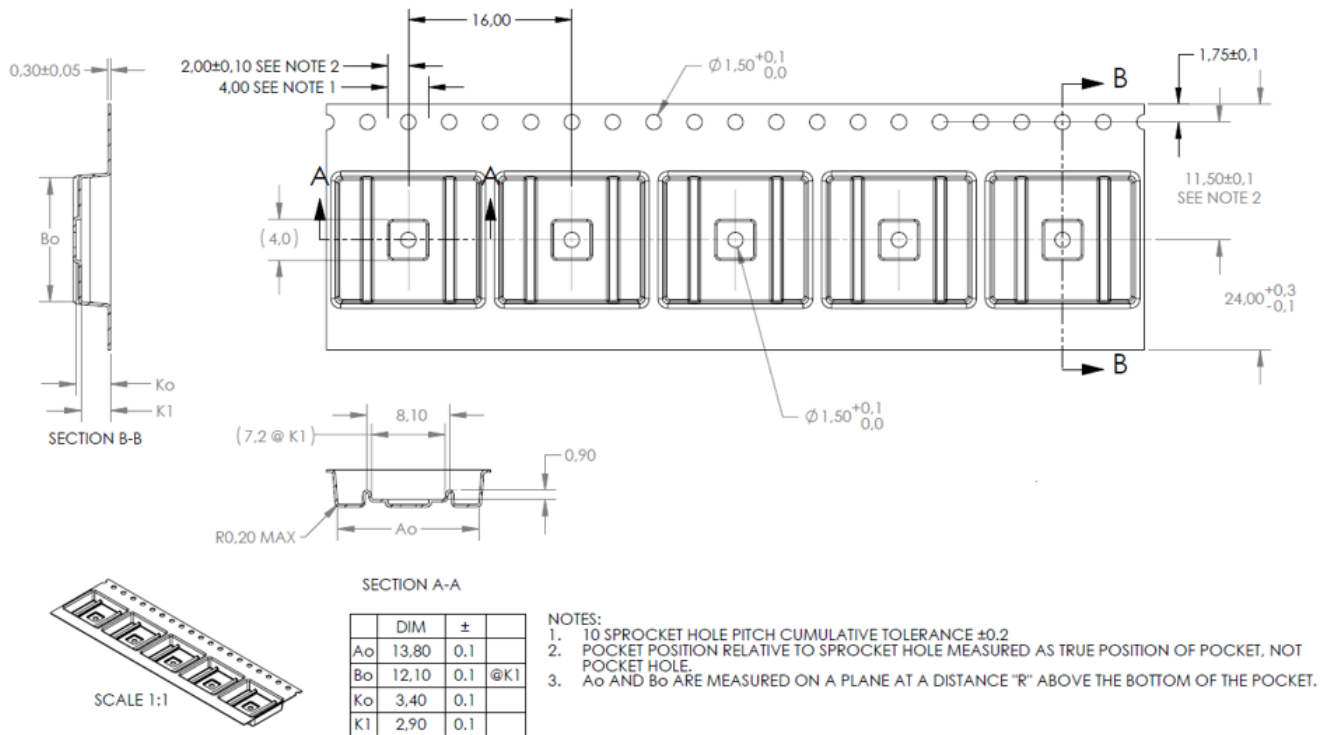
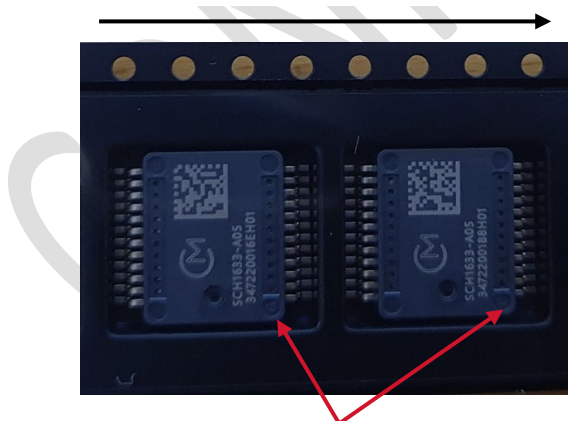


Figure 3 Packing tape specification. Dimensions are in millimeters [mm]

Direction of unreeling



Pin 1

Figure 4 Component position in carrier tape

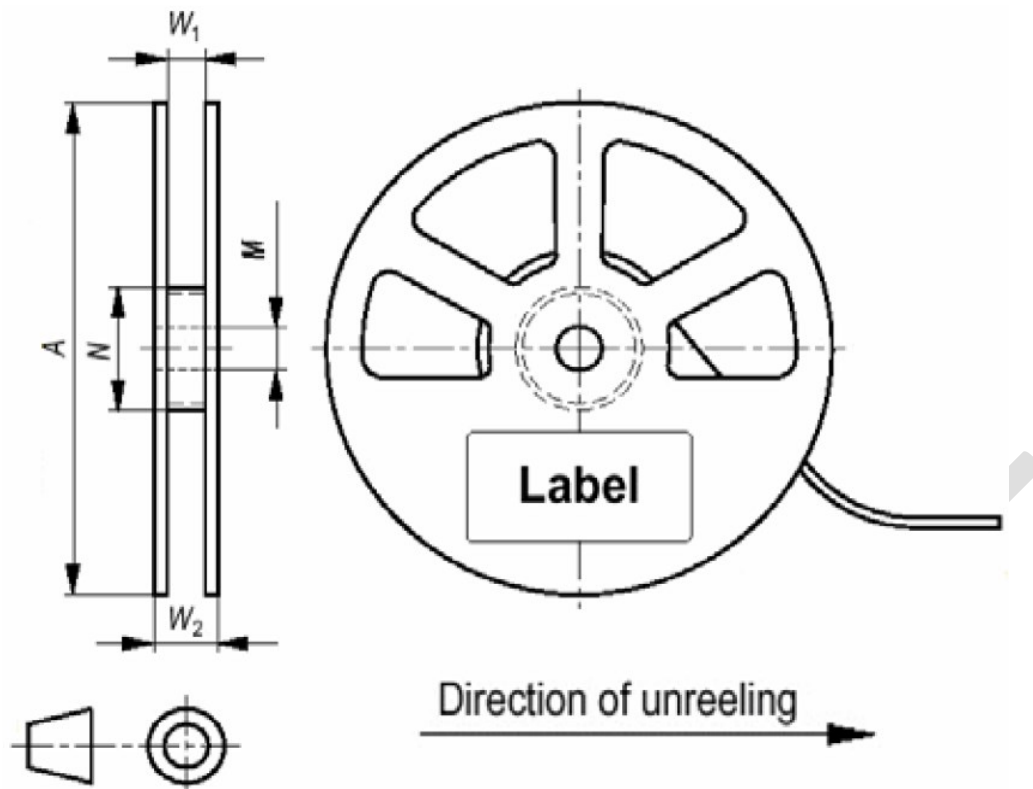


Figure 5 General figure of packing reel

Table 1. Packing reel dimensions [mm]

A	N	W1	W _{2max}	M
330	100±1	32.4 (-0/+2.0)	38.4	∅ 13 (-0.2/+0.5)

5 Printed Circuit Board (PCB) design and assembly guidelines

5.1 PCB terminal pad design

PCB terminal pads should be wider than package lead to improve solder joint quality and reliability. A reference PCB terminal layout is presented in **Figure 6**. Note that pad design can be customized to suit PCB requirements. (Recommendation).

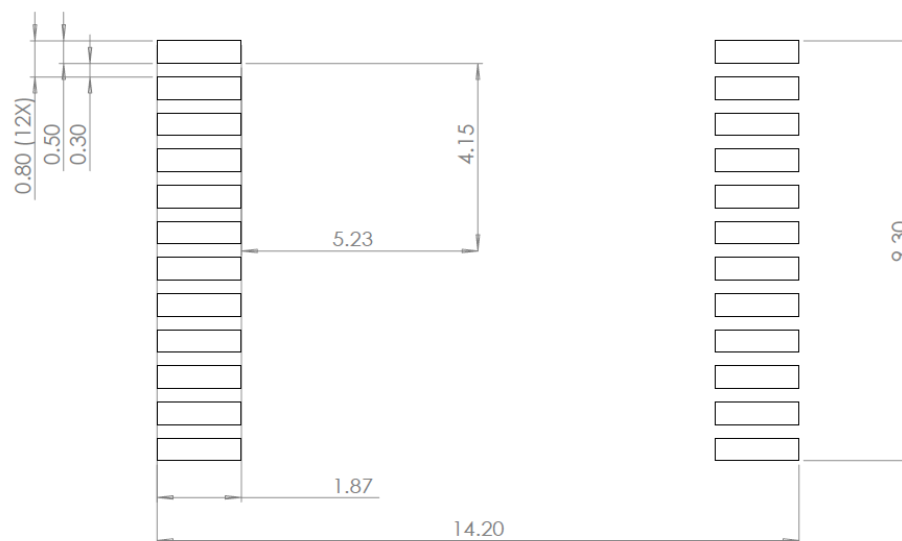


Figure 6 Reference pad layout

Pad metallization must be solder wettable to assure good quality solder joints. Recommended circuit board finishes for SMD soldering are NiAu, OSP, electroless-Ag and electroless-Sn.

5.2 Passive component placement under component legs

Due to the SCH1000 component design, space is left between the component legs and PCB. If it is considered feasible for the PCB design and assembly line, this space may be utilized for passive components. Murata has verified concept with simulations and test PCBs but does not take responsibility of assembly or operation failures related to placement of passive components under component. Effect of component gluing and/or coating and using thermal pads with passives underneath component legs must also be evaluated separately.

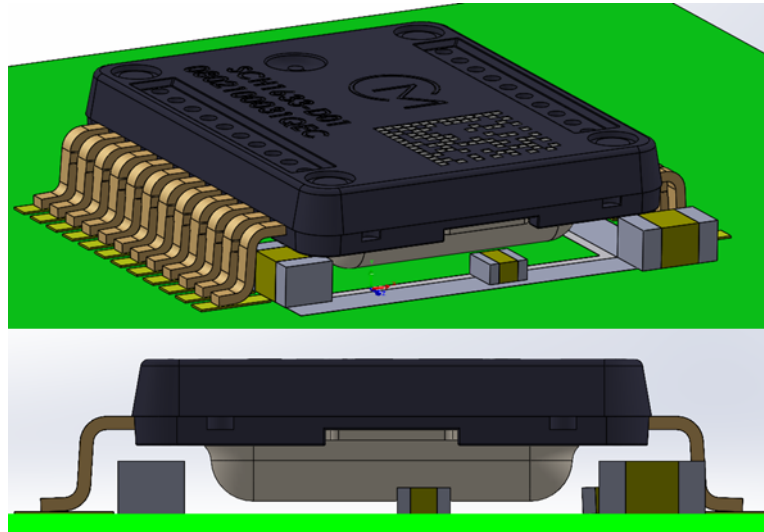


Figure 7 Schematic of passive component placement concept

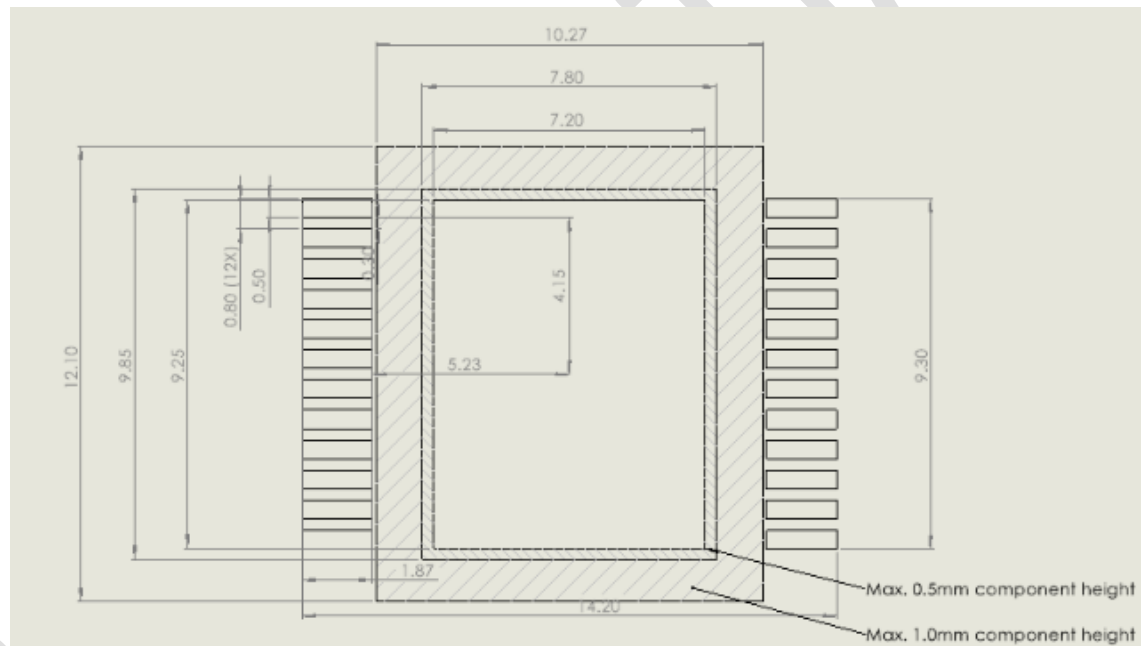


Figure 8 Area available for passive components. All measurements are in millimeters [mm]

If passive components are soldered under the SCH1000 component it must be done prior to soldering of the component. The passive components must be also considered in pick and place process as per chapter 5.9 (Requirement).

Placing passive components under the SCH1000 may affect the ability to make solder quality control. Using X-ray to inspect solder quality may be necessary (Recommendation).

5.3 Temperature effects

5.3.1 General

Arrangement of mounted parts on front and back surfaces and their coefficients of thermal expansion (CTE) should be considered to reduce PCB warpage. Conductor occupancy for front and back side of the PCB should be similar to minimize PCB warping. Heat resistant board types should be used to reduce warpage. Prevention mechanisms, such as PCB fixture tools can be used to reduce PCB warping during reflow. (Recommendation).

Heating effect of other components shall be considered when designing the PCB. It is not recommended to expose the SCH1000 component to uneven heating. When using multiple SCH1000 components, it is recommended that all components would experience a similar heat load to retain the greatest accuracy. This is especially important if the data from multiple components is used in comparison with each other. (Recommendation).

5.3.2 Thermal conductor

A thermal pad may be applied underneath the component to improve heat flux. A thermal pad may also reduce housing resonance accelerations and shift housing resonances to higher frequencies.

- Thermal pad material
 - The pad material shall not be conductive to avoid risk for short circuit between component legs.
 - The pad should not contain prohibited contaminants.
 - Murata does not provide recommendations for thermal pad material.
- Pad geometry
 - The pad must be applied before soldering process. The pad height should be chosen so that the paste will not spread into contact with legs, but still provides adequate and even contact area on the bottom lid.
 - The thermal pad should be axially symmetrical and have an even height.
- Pad effect on component performance
 - User is responsible to validate functionality and performance of sensor if a thermal pad is used (Requirement)
 - See also chapter Precautions

Customer must confirm reliability of their thermal pad applying process by adequate reliability testing (thermal pad may be subjectable to lifetime effects). Care must be taken that system resonance frequencies are not shifted to prohibited frequency bands. For further instructions about vibration reduction, please refer to chapter 5.5.

5.3.3 External heating

If external heating and/or cooling is applied to the component, all temperature effects mentioned previously in this chapter must be considered. The effect of external heating/cooling to component must be assessed separately.

5.4 De-paneling

PCB de-paneling is a source of mechanical or vibrational stress to the SCH1000 component. PCB panel breakout tabs should be designed with minimum width and SCH1000 components should not be placed near the tabs. Holes on breakout tabs are not recommended when cutter, saw, or router is used for de-paneling, because the holes can increase the number of vibrational spikes. (Recommendation).

5.5 Vibration reduction on PCB level

Performance might be inferior if the application PCB induces high resonance gain to the sensor component. Therefore, properly designed PCB attachment points together with PCB layout is of utmost important for applications that are intended to be used at vibrating environment.

5.5.1 Vibration mechanical range in application and in assembly process

It is recommended that user measures the resonance gain over frequency from the vicinity of the SCH1000 component at the application and in assembly process. In order to maintain proper operation of the sensor, the g-values at the application use environment must not exceed the specified mechanical range of the sensor component (refer to the component datasheet and chapters "Vibration reduction on PCB level" and "Precautions" of this document) (Requirement).

For example, PCB resonance modes can amplify external harmonic vibrations that can cause excessive vibration input to sensor component that exceeds the specified mechanical range. Murata cannot guarantee operation in conditions where the mechanical range of SCH1000 component is exceeded and takes no responsibility if sensor operation failure occurs in such cases.

5.5.2 Application and assembly process induced frequencies

Sensor component can have adverse performance if the application or assembly process induces vibration peaks at critical frequency bands of the sensor component. It is advised that application and assembly process induced vibrational peak frequencies at the vicinity of the SCH1000 component should not overlap with Table 2 frequencies. Such vibration peaks can be caused by e.g., natural modal resonances of the use application's structures (PCB, lid, casing, mechanical fixtures) or other components at the use environment.

Table 2 Murata recommends to design PCB and system so that they don't have resonance frequencies in these frequency bands (Recommendation). Vibration at these frequency bands may be visible in component output.

Frequency bands to avoid by component linear excitation direction		
X	Y	Z
9.5 kHz $\pm$ 1 kHz	8.6 kHz $\pm$ 1 kHz	12.2 kHz $\pm$ 1 kHz
	23.6 kHz $\pm$ 1.5 kHz	
27.8 kHz $\pm$ 1 kHz	-	-
	37 kHz $\pm$ 2 kHz	

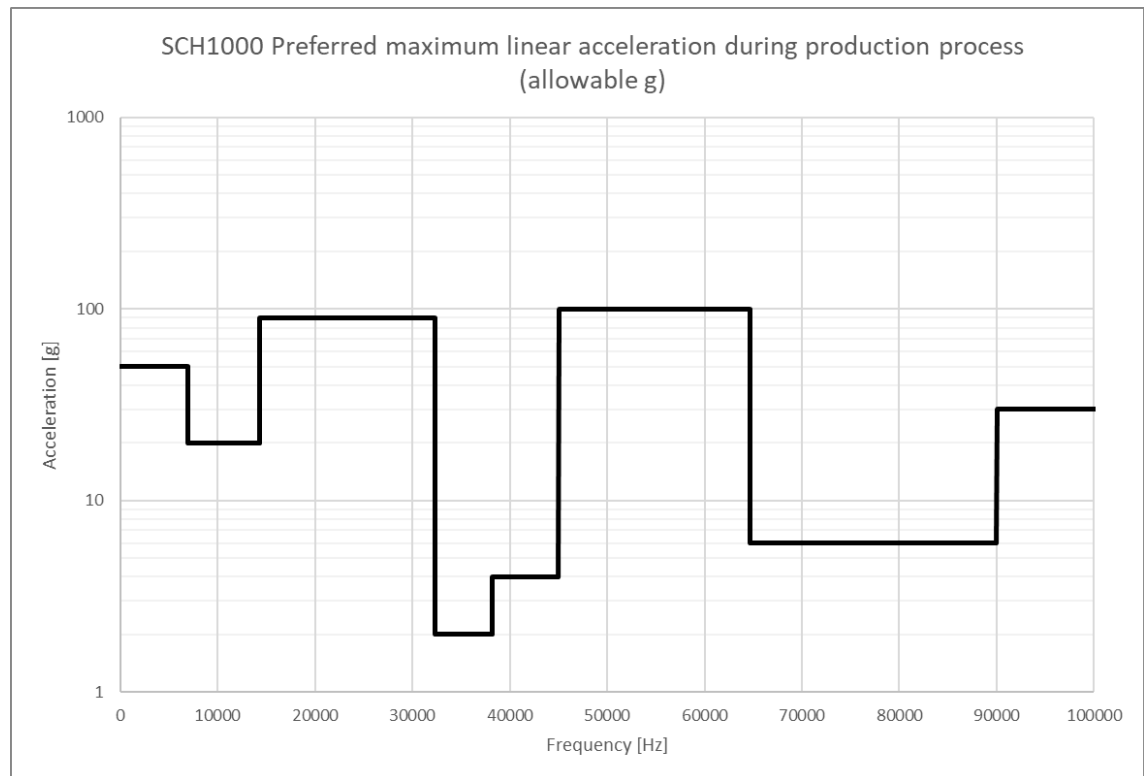


Figure 9 Preferred maximum linear acceleration during production process based on simulations and verified with practical tests. Prolonged vibration exceeding this curve may cause damage to sensor (Murata has verified that 10 g @ 5 kHz-50 kHz vibration with 2000 Hz/min sweep doesn't damage the sensor). This curve contains safety margin, but Murata recommends operating under the accelerations specified in this graph. (Recommendation).

5.5.3 Vibration reduction by PCB Design

PCB design has a major impact on system level performance for applications that are intended to be used at vibrating environment.

PCB properties

- Symmetricity
 - Murata recommends using symmetrically shaped PCB to minimize the potential number of vibrational modes on system level.

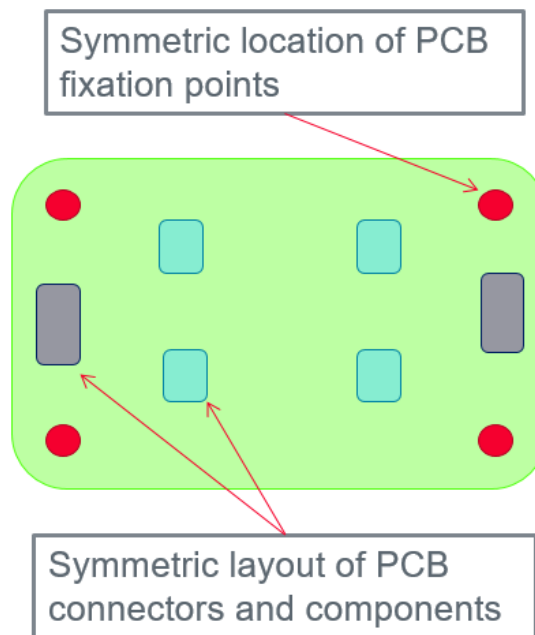


Figure 10 PCB symmetry

PCB attachment

- The distance from attachment points to the component (e.g. free span) affects stiffness, fundamental frequency and damping properties of the PCB that the component experiences. In general, the center part of the PCB is more exposed to Z-directional acceleration than PCB edges, therefore recommendation is to prefer PCB edge areas for inertial sensor location than center area.

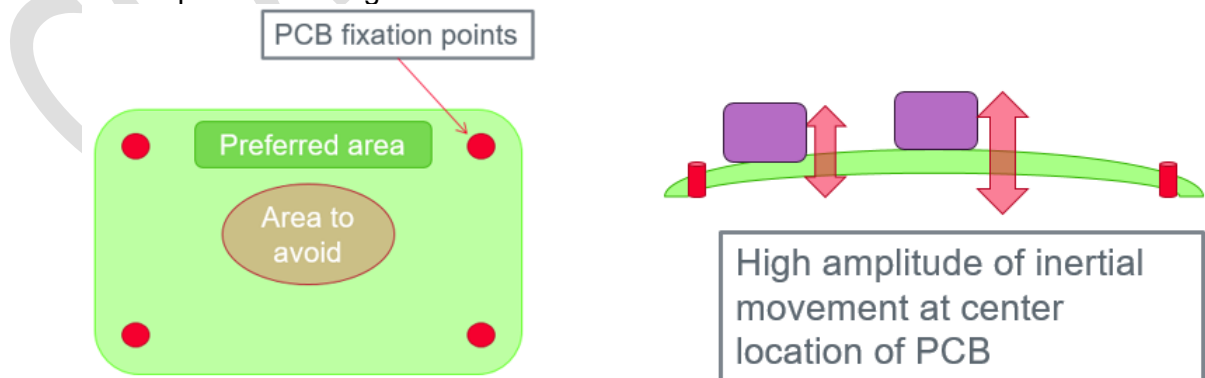


Figure 11 PCB attachment point distance effect

- Attachment method and the amount and location of attachment points have a major effect on the PCB behavior.
 - Increased amount of attachment points in preferably symmetric locations stiffens the PCB and has potential for improving vibration behavior.
 - It is recommended to minimize free span over inertial sensor component by adding extra fastening points around sensor component.
 - Secure and tight fastening of PCB to system minimizes the movement of PCB and excitation of vibration modes.
 - Always use high stiffness attachment method like thick screws or similar, to attach the PCB to next level system. Using of loose attachment method, e.g., press fit contacts, for mechanical mounting of PCB will increase risk of high amplitude PCB vibration modes that can cause sensor operation failure.



Figure 12 PCB attachment point location effect

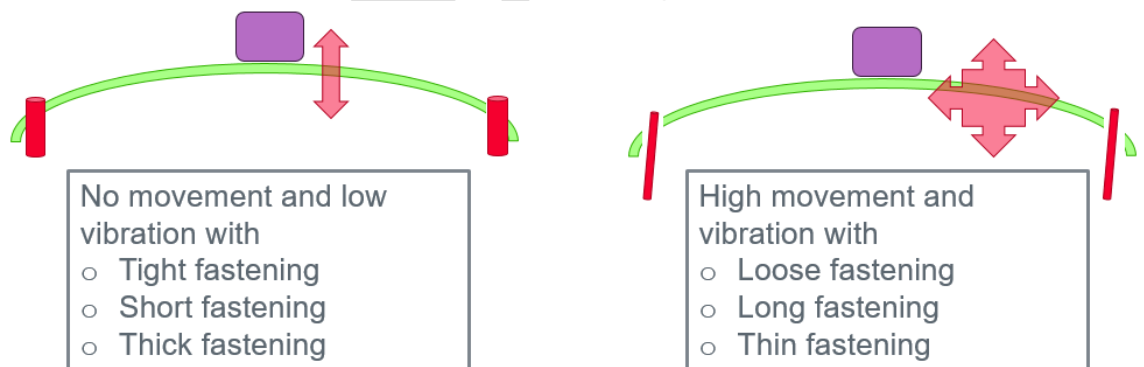


Figure 13 PCB attachment point type effect

PCB size and thickness (cross-sectional area)

- Cross-sectional area affects stiffness, fundamental frequency, and damping properties of the PCB. In general
 - Smaller PCB size will have lower Z-directional amplitude and higher vibrational frequencies.
 - Thicker PCB will increase vibrational frequencies.
 - Number of copper layers stiffens PCB and increases vibrational frequencies.
- PCB size and thickness should be chosen so that the PCB resonances will not overlap with Table 2 frequencies.

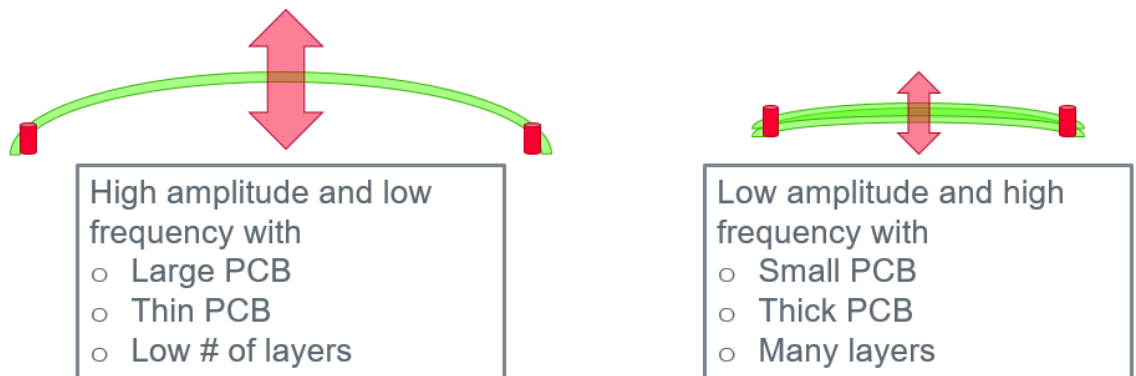


Figure 14 PCB size and thickness

PCB material

- Elastic modulus affects stiffness, fundamental frequency, and damping properties of the PCB. In general, high modulus material will
 - increase PCB stiffness.
 - increase vibrational frequencies.
 - decrease damping properties of PCB.
 - If the PCB is too flexible, it can be artificially stiffened using PCB stiffening ribs.

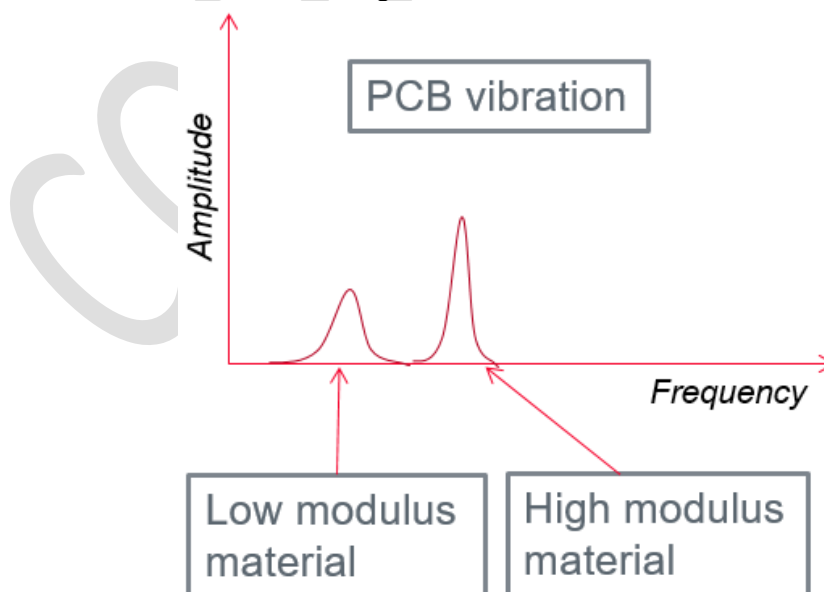


Figure 15 PCB material property effect to vibration performance

PCB attachment

- Rubber dampers/vibration mounts at the attachments can reduce high frequency vibrations.
 - It is possible to use rubber dampers at module level also.
 - Good shock isolator is not always a good vibration isolator due to dynamic strain properties of elastomers.
- If PCB is attached using screws, the screw material and screw length has an effect on the vibration behavior.
- PCB edge guides have a slight effect on the vibration behavior.

Effect of other components

- To avoid dynamic coupling, the resonances of other components near SCH1000 components should be at least one octave away from the resonance bands defined in Table 2
 - If it is not possible to avoid these components or resonances cannot be tuned, the components should be laid out as far away from each other as possible.
- High mass components should be laid out on the PCB symmetrically.

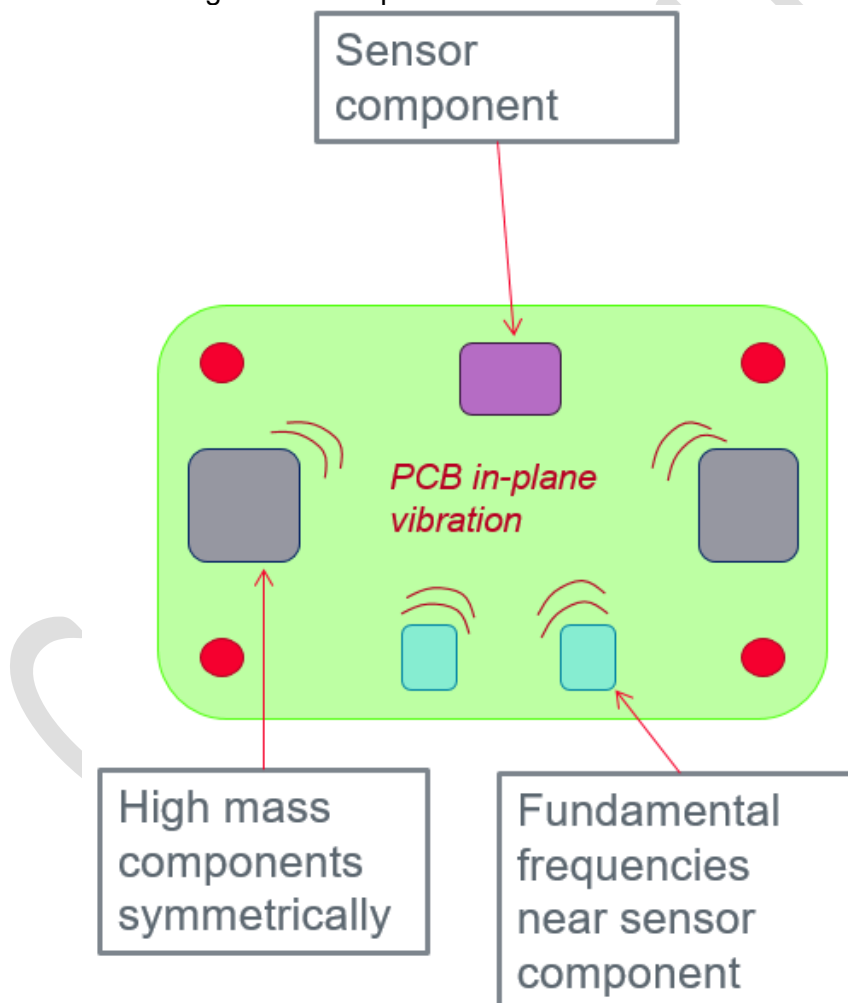


Figure 16 Effect of other components

5.5.4 Conformal coating

Conformal coating reduces housing resonance accelerations and shifts housing resonances to higher frequencies. If conformal coating is used for vibration reduction, several effects should be considered:

- Coating thickness
 - A thick coating usually reduces vibrations better than a thin coating.
- Coating material
 - Polyurethane and silicone based conformal coatings are known to be good materials for vibration reduction.
 - When choosing a coating material, the most important properties in vibration point of view are coating's ability to form a thick layer and elastic modulus of the coating.
 - Coating material should not contain prohibited contaminants.
- Symmetricity and area of the coating
 - There is a major difference if the whole component is coated vs body only/legs only.
 - Coating should be axially symmetrical i.e., spraying process should be well controlled.
- Coating effect on component performance
 - User is responsible to validate functionality and performance of sensor if conformal coating is used (Requirement)
 - See also chapter Precautions

Customer must confirm reliability of their conformal coating by adequate reliability testing. Customer should avoid shifting system resonance frequencies to the critical frequency bands shown in Table 2. (Recommendation).

5.5.5 Gluing

Due to the component design, gluing component body to the PCB is not recommended. The metal lid on the bottom of the component may experience stress from the glue, especially if exposed to a temperature gradient. This may in worst case lead to detachment of lid, component, or glued interface. Component legs can be glued after the SMT solder process by dispensing a large amount of glue on the legs.

If gluing is used for vibration reduction despite the recommendation, several effects should be considered:

- Glue material
 - Softer materials usually reduce vibration more than rigid materials (difference in elastic modulus)
 - All glues do not have adequate adhesion to component body or PCB.
 - The glue should not contain prohibited contaminants.
- Glue geometry
 - If glue is dispensed before soldering process, the glue dot height should be chosen so that the glue will not spread into contact with legs (but still providing adequate contact area on the component body)

- Murata does not recommend using only one large glue drop under component.
- Gluing should be axially symmetrical. If multiple glue dots are used, the dot height and dot positioning should be even.
- Underfill type of hard full bottom area attachment is not recommended.
- Gluing effect on component performance
 - User is responsible to validate functionality and performance of sensor if component gluing is used (Requirement)
 - See also chapter Precautions

Customer must confirm reliability of their gluing process by adequate reliability testing (gluing may brake over lifetime). Care must be taken that system resonance frequencies are not shifted to prohibited frequency bands.

5.6 Solder paste

Recommended solder paste for the SOIC package is near eutectic lead-free SAC (tin-silver-copper) solder with melting point between 217–221°C.

A no-clean solder paste is required as washing the component is not allowed.

Ultrasonic agitation is strictly prohibited for Murata's MEMS components since it can destroy the MEMS structures (Requirement).

5.7 Stencil and paste printing

It is recommended to apply solder paste onto the PCB using stencil printing. Minimum stencil thickness that can be applied is 0.125 mm (Recommendation). Customer should validate system level reliability and vibration performance with their method of solder paste printing.

5.8 Paste printing

The paste printing speed should be adjusted according to the solder paste specifications. It is recommended that proper care of printing speed is taken during the paste printing to ensure correct paste amount, shape, position, and other printing characteristics. Neglecting any of these can cause open solder joints, bridging, solder balling, or other unwanted soldering results.

5.9 Component picking and placement

Typically, SOIC package is picked from the carrier tape using vacuum assist type pick heads. It is recommended to test different pick-up nozzles to ensure accurate placement of the component.

Placement should be done with modern automatic component pick & place machinery using vision systems. Recognition of the packages automatically by a vision system enables correct centering and orientation of packages.

Force during placement of the component must be adjusted to not cause damage to the component. Additionally, if passive components are placed under the component according to proposition in chapter 5.2, the trajectory of the pick and place nozzle and placement force and accuracy must be optimized to not damage these components.

5.10 Reflow soldering

A forced convection reflow oven is recommended to be used for soldering SOIC components. IR-based reflow ovens are not generally suitable for lead-free soldering. Figure 7 presents a general forced convection reflow solder profile and it also shows the typical phases of a reflow process. The reflow profile used for soldering the SOIC package should always follow the solder paste manufacturer's specifications and recommended profile. Note that washing of the component is not allowed (Requirement).

Too high a reflow peak temperature can have an adverse effect on the sensor performance. The peak reflow temperature measured from the top of the package body should not exceed 250°C and the time at the package peak temperature of 250 +0/-5°C should not exceed 30 seconds (Recommendation). The absolute maximum package peak temperature of SCH1000 component is 260°C and time within the temperature of 260 +0/-5°C must not exceed 30 seconds (Requirement). Murata cannot guarantee operation in the case when the package peak temperature specification is exceeded and takes no responsibility if sensor operation failure occurs in such a case.

Maximum number of reflow cycles for SCH1000 components is three (Requirement).

Reflow soldering with vacuum condition is not guaranteed by Murata. If reflow soldering with vacuum conditions is used, customer should confirm reliability of the reflow process by adequate reliability testing. (Recommendation).

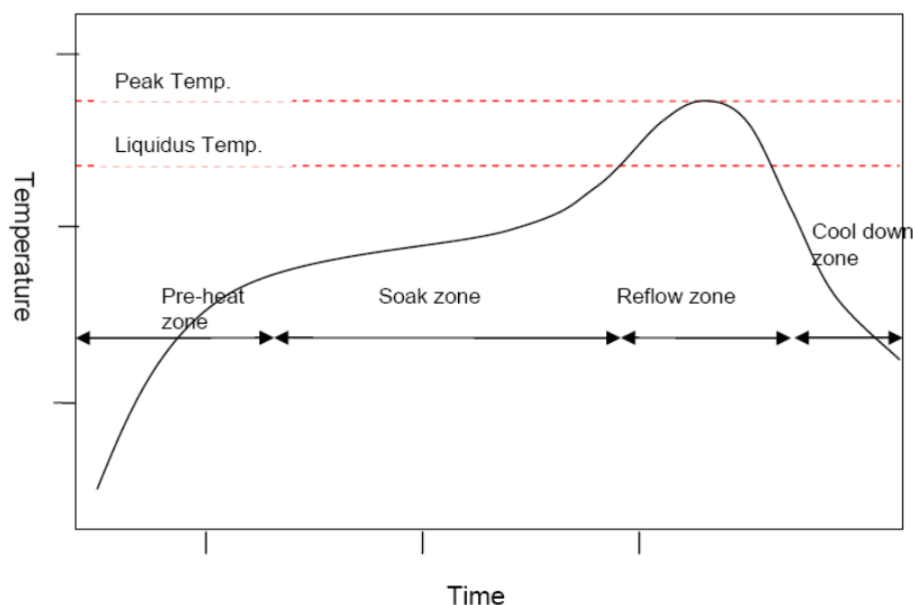


Figure 17 Typical convection reflow soldering phases and profile

PCB boards can have a large temperature variance over the board area during reflow due to uneven thermal mass distribution and oven properties. User should measure package peak temperature (peak reflow temperature) and solder joint temperature profiles at various locations on the board to ensure correct conditions for package integrity, solder joint formation, and component self-alignment.

Reflow profile measurements should be made with thermocouples following the guidelines described in *JEDEC JEP140: Beaded Thermocouple Temperature Measurement of Semiconductor Packages*. Package peak temperature is measured from the center top surface of the component and solder joint temperature is measured directly from the solder joint contact. Fixing of a thermocouple should be made with minimum amount of thermally conductive attachment medium. Large measurement error may result if the thermocouple is fixed with media having low thermal conductivity, such as polyimide tape. Temperature profiles should be adjusted across the whole PCB to minimize temperature gradient across the circuit board and to provide proper soldering temperature profile of all solder joints. Extreme caution must be used if the circuit board contains components with highly different thermal masses. (Recommendation).

5.11 Moisture sensitivity level (MSL) classification

The Moisture Sensitivity Level of the SOIC component is Level 3 according to the IPC/JEDEC J-STD-020E. The parts are delivered in a dry pack.

Following instruction shall be followed:

1. Calculated shelf life in sealed bag: 12 months at $< 40^{\circ}\text{C}$ and $< 90\%$ relative humidity (RH) (Requirement).
2. Maximum package peak temperature for the package is 260°C , maximum package peak temperature time is 30sec within $260 \pm 5^{\circ}\text{C}$, measured from the package top surface (Requirement).
3. After bag is opened, devices that will be subjected to reflow solder or other high temperature process must be
 - a) Stored at $< 10\% \text{RH}$
 - or
 - b) Mounted within 168 hours of factor conditions $\leq 30^{\circ}\text{C}/60\% \text{RH}$. (Requirement)

Note: Do not re-store devices that have exposed $> 10\% \text{RH}$ conditions.

4. Devices require bake, before mounting, if:
 - a) Humidity Indicator Card is $> 10\%$ when read at $23 \pm 5^{\circ}\text{C}$
 - b) 3a or 3b not met. (Requirement)
5. If baking is required, minimum baking time is 96 hours at $85^{\circ}\text{C} \leq 5\% \text{RH}$ (Requirement).

Note: Also Tape&Reel materials are applicable for baking at 85°C .

6. Long-term storage of components shall always be within sealed dry bag (Requirement).
7. To guarantee lifetime targets following storage conditions for sealed vacuum dry packed components shall be applied: $+15^{\circ}\text{C}$ to 40°C , humidity $\leq 60\% \text{RH}$ (Requirement).

8. The buildup of impureness on soldering pins from atmosphere can affect solderability. The storage time without vacuum package should be minimized. The maximum storage time without vacuum package must not exceed 6 months (Requirement).
9. Maximum storage time before soldering: 1 year (Recommendation), 2 years (Requirement).

Note: Packing materials and procedures according to IPC/JEDEC J-STD-033D

Note: Level and body temperature defined by IPC/JEDEC J-STD-020E

5.12 Inspection

Optical and visual inspection of solder joints can be done easily, because the solder joints are clearly seen. A visual inspection of the solder joints with conventional AOI (automatic optical inspection) system can be used. Also X-ray inspection can be used.

Cross-sectional analysis is also an approved method to inspect how well solder has wetted the pads of component. Cross-sectional analysis is not used for production inspection, but if required, it can be used to establish and optimize the component assembly process parameters.

5.13 Precautions

MEMS sensors are mechanically and electrically sensitive components. Following sections describe precautions for sensor component. Exceeding any of these limits or neglecting these guidelines may lead to immediate malfunction of the sensor component or malfunction that occurs after a period. Murata does not take any responsibility if there has been any violation against these precautions.

The reliability requirements for the devices are applied and validated according to AEC-Q100 Rev. H.

5.13.1 Mechanical shocks and vibration during handling and assembly

Shocks may cause mechanical damage to the internal structures of MEMS sensor, causing malfunction of sensor, therefore mechanical shocks should be avoided. The level depends heavily on the pulse width and shape and should be evaluated case by case. As a general guideline, the lighter assembly or part, the higher shock levels will be generated on sensor component.

Any dropped component shall not be used and shall be scrapped (Requirement).

Sensor components are mechanical devices and especially sensitive to repetitive vibrations and shocks, therefore vibration of the device should be avoided both prior to and during assembly. Many assembly processes can induce vibration, typical ones being PCB singulation, dropping or knocking parts on hard surfaces, transportation, friction welding, ultra-sonic cleaning, and usage of pneumatic wrenches.

5.13.2 Chemicals

Sensor components shall not be exposed to (Requirement):

- Chemicals which are known to react with silicones, such as solvents. These are for example used in various cleaning processes.

- Chemicals with high impurity levels, such as Cl⁻, Na⁺, NO₃⁻, SO₄⁻, NH₄⁺
- Pressurized low molecular gas such as He or H₂, used for example in leak test for hermetical sealing.
- Materials with high amount of volatile content, like solvents.

Materials containing halogens (F, Br, I or Cl), halides, their compounds or phosphorus containing materials (such as in flame retardants, thermal stabilizers in plastics, Bromine in PCB material or halogens/halides in soldering paste) shall be avoided in close vicinity of sensor component. (Requirement).

If heat stabilized polymers are used in application, user should check that iodine, or other halogen, containing additives are not used. Halogen free solder paste classified by IEC61249-2-21 may have Iodine additive instead of limited Cl and Br in the IEC standard (Br<900ppm, Cl<900ppm and Br + Cl <1500ppm). Iodide compounds are known to cause corrosion issues with gold aluminum interconnects. User should also check that excessive amount of I, Br or other halogen containing additive are not used (Requirement: zero iodine, no intentional use in user application). Quantitative value is depending on number of factors, including the total volume of such halogen material, stability of halogen within material, hermeticity level of application and temperature, among others. (Requirement).

Life-time reliability tests should always be performed at application level to validate the end product against life-time requirements/mission profile, as corrosion effects heavily depend on the final construction of application, temperature, time and other application specific factors.

5.13.3 Coatings and nano-coating

User has the responsibility to validate use of any material or structure that modifies mechanical connection between the sensor and the PCB, e.g., coating, lacquering, or gluing to PCB. (Requirement). However, Murata has verified that a nano-coating type coating does not affect the component over temperature or lifetime performance.

5.13.4 Vacuum level

Performance of SCH1000 components at high vacuum conditions is not guaranteed by Murata. User is responsible to validate functionality and performance of sensor if it is exposed to or used under high vacuum levels. (Requirement).

5.13.5 ESD

Sensor components are electrical devices. Sensor components should be handled under good ESD practices and ESD discharges should be avoided. The following numbers are absolute maximum ratings:

- ±500 V charged device model, ±750 V for corner pins device model (Requirement).
- ±2 kV human body model (Requirement).

5.13.6 Moisture

Sensor components are moisture sensitive devices, classified as MSL3 level. Guidelines defined by PC/JEDEC J-STD-033 shall be followed (Requirement).

5.13.7 Mechanical stress

Mechanical stress due to PCB bending, molding, coating, and potting may affect the sensor performance. Excessive stress due to PCB bending shall be avoided (Recommendation). User is responsible to validate functionality and performance of sensor if molding or potting material is used (Requirement).

PCB bending over the sensor area should be limited to 1.0% (Recommendation). User is responsible to validate functionality and performance of sensor due to PCB bending (Requirement).

5.13.8 Cleaning

Ultrasonic cleaning is strictly prohibited (Requirement).

Items listed in "Precautions" section are applicable for cleaning. In addition to ultrasonic cleaning other types of cleaning might have negative effects on the component performance or functionality. Murata recommends validating following cleaning procedures (Recommendation):

- Wet cleaning with solvent
- High velocity or pressure air blowing
- Plasma cleaning with vacuum

5.13.9 Magnetic field

Avoid operation the sensor in a strong magnetic field (Recommendation).

6 Rework Guidelines

If it becomes necessary to detach a SCH1000 component from a PCB, the preferred way to remove the component is by hot air. The part together with the PCB must be dried in an oven at 95°C +/- 5°C for 48 hours +5/-0 hours and detachment must be made within 168 hours after the drying is completed (Recommendation).

A detached component cannot be re-used and must be scrapped (Requirement).

7 Environmental Aspects

Murata Electronics Oy respects environmental values and thus, its SOIC packages are lead-free, Halogen free and RoHS and China RoHS compatible. Murata Electronics' sensors should be soldered with lead-free solders to guarantee full RoHS compatibility (Recommendation).

8 References

JEDEC / Electronic Industries Alliance, Inc. Moisture/Reflow Sensitivity Classification for Non-Hermetic Solid State Surface Mount Devices (J-STD-020E).

Handling, Packing, Shipping and Use of Moisture/Reflow Sensitive Surface Mount Devices (IPC/JEDEC J-STD-033).

JEDEC Solid State Technology Association. Beaded Thermocouple Temperature Measurement of Semiconductor Packages (JEDEC JEP140).

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